UGANDA REVISION HUB

S2 Starter Study Notes

Batch 1 • Online and offline revision

S2 • Mathematics

Algebraic expressions

Learning objectives

- Identify variables, coefficients and constants
- Collect like terms
- Substitute values into expressions

Lesson notes

An algebraic expression combines numbers, variables and operations. A variable is a letter representing a number, such as x. In 5x + 3, 5 is the coefficient of x and 3 is the constant.

Like terms have the same variable raised to the same power. They can be combined: 3x + 2x = 5x, but 3x + 2y cannot be combined into one term.

To evaluate an expression, replace each variable with the given value and calculate using the correct order of operations.

Worked example

Simplify 4a + 3a - 2 = 7a - 2. If a = 2, the value is 14 - 2 = 12.

Revision questions

1 Simplify 6x + 2x - 5.

2 Evaluate 3y + 4 when y = 5.

3 Explain why 2a and 2b are not like terms.

Suggested answers

1 8x - 5

2 19

3 They contain different variables.

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S2 • Biology

Nutrition in plants

Learning objectives

- Define photosynthesis
- State the raw materials and conditions for photosynthesis
- Write the word equation for photosynthesis

Lesson notes

Photosynthesis is the process by which green plants use light energy absorbed by chlorophyll to make glucose from carbon dioxide and water, releasing oxygen.

Carbon dioxide enters leaves mainly through stomata. Water is absorbed by roots and transported through xylem. Chlorophyll in chloroplasts absorbs light energy.

Photosynthesis needs light, chlorophyll, carbon dioxide and water. Glucose may be used in respiration, stored as starch, or used to make other substances.

Worked example

Carbon dioxide + water —(light and chlorophyll)→ glucose + oxygen.

Revision questions

1 Name the green pigment needed for photosynthesis.

2 Where do plants obtain carbon dioxide?

3 Why is light needed for photosynthesis?

Suggested answers

1 Chlorophyll

2 From the air, entering mainly through stomata

3 It provides energy for the process.

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